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Trauma-focused CBT with maltreated children: A clinic-based evaluation of a new treatment manual

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Abstract
The effectiveness of a locally developed trauma-focused cognitive behavioural therapy (TF-CBT) program for maltreated children with post-traumatic stress disorder (PTSD) was examined across different therapists in a child protection clinic setting. An earlier phase of the research piloting the program had provided promising results. This second phase involved two studies evaluating the completed TF-CBT manual delivered by (a) the developer and (b) other therapists. A single-case multiple-baseline design was used to demonstrate the controlling effects of the treatment on PTSD symptoms and child coping. Eight 9–13-year-old abused children with PTSD were treated. Positive outcomes support the effectiveness of the TF-CBT program delivered by both the developer and other therapists. The study design and methodology were robust enough to confirm empirically the clinically beneficial effects and potential for this new program. It was also apparent, however, from study limitations, including missing data for some patients, that there are a number of challenges in carrying out such research in a busy child protection service setting with multiply-abused patients. This paper considers implications and ways forward for engaging in empirically supported practice as well as future development and research.

Key words: *Child and adolescent psychopathology, child maltreatment, clinical research, clinical research methods, clinical/counselling psychology, post-traumatic stress disorder.*

Recently there has been emerging recognition of the need for evidence-based treatments in the field of child maltreatment (Chaffin & Friedrich, 2004; Macdonald, 2001; Skowron & Reinemann, 2005). This is a relatively new idea for this field and is based on clinical science (Feather, 2004) rather than on clinical lore or traditions. The current research takes the view that in order to ensure safe and effective outcomes for maltreated children and their families in the local clinical setting, local clinical practice is best interwoven with well-supported treatment models that have scientific evidence for proven outcomes. This article presents studies from a treatment development project designed to develop and evaluate a manualised trauma-focused cognitive behavioural therapy (TF-CBT) program for maltreated children diagnosed with post-traumatic stress disorder (PTSD) (Feather & Ronan, 2004). The main intention of this project is to develop and evaluate a manualised treatment program that clinicians can use to help maltreated children build skills to manage their symptoms and process the trauma of abuse, with the support of their parent or caregivers. This TF-CBT program builds on efficacious treatments for child anxiety and PTSD as a result of sexual abuse.

International child and adolescent treatment outcome research has found CBT to be efficacious for a range of child problems. Efficacy and effectiveness studies support the usefulness of CBT for anxiety disordered children (Barrett, 1998; Barrett, Duffy, Dadds, & Rapee, 2001; Howard & Kendall, 1996; Kane & Kendall, 1989; Kendall, 1994; Kendall & Southam-Gerow, 1996), and TF-CBT for children with PTSD as a result of child sexual abuse (Cohen & Mannarino, 1996, 1997; Deblinger, Lippmann, & Steer, 1996; Deblinger, Steer, & Lippmann, 1999; King et al., 2000) and multiple abuse (Cohen, Deblinger, Mannarino, & Steer, 2004). CBT approaches highlight the importance of a collaborative therapeutic relationship, as well as use of creative applications of the model, particularly when working

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with children (Kendall et al., 1992). These ideas underpin the aim of the therapist and child (and parent/caregivers) working together with a range of modalities to alleviate symptoms and enhance coping such that the child will think, feel and behave differently in the present and the future, beyond the experience and effects of abuse (Kendall, 2000).

Developing a treatment for maltreated children involves special considerations. These include the essential need to take a developmental and culturally informed perspective, and a family and wider systems approach (Feather, 2007; Jackson, Frederico, Tanti, & Black, 2009). Child maltreatment almost always affects the relationships that children have with their family members. In addition, many children who come to the attention of child protection services are placed in care to ensure their safety. Removal from parents may add another layer of trauma for these children, and necessitates forming new relationships with caregivers as well as social service professionals. As a consequence, the early part of this treatment program is devoted to exploring and strengthening the child's psychosocial context as a basis for the later treatment interventions. Throughout the program, parents and caregivers are included, a choice of child activities is provided to cater for different developmental needs, and recommendations are made to ensure that the approach is culturally respectful. Abuse-informed issues are incorporated in the treatment, as recommended in practice guidelines (Chadwick Center for Children and Families, 2004; Saunders, Berliner, & Hanson, 2004). The program is presented in four parts, or phases, based on practice wisdom, theory and empirically supported treatments.

The coping skills phase of the TF-CBT program was based on the "Coping Cat" program for anxiety disordered children developed by a team that included the second author (Kendall et al., 1992; Ronan & Deane, 1998; see also Kendall, Kane, Howard, & Siqueland, 1990). A key consideration in adapting an anxiety-based program to the treatment of PTSD is that many forms of anxiety are about current and future threat whereas PTSD, while an anxiety disorder, also has much to do with a past event(s). Children with PTSD are processing trauma and/or its sequelae in a way that often involves not only future-oriented worries but also ongoing and distressing reminders (Ehlers & Clark, 2000). Hence, the aim of a treatment program for PTSD needs to be to help children develop skills to manage their symptoms, and to process trauma so that it is seen as a time-limited past event(s) that can be managed effectively by the child and their family/caregivers, coupled with a focus on current and future concerns.

Supported by the evidence, a CBT approach holds that gradual exposure is an optimal strategy for reducing PTSD symptoms (Kendall et al., 1992). This can be achieved in a number of ways, including using creative media to create a coherent trauma narrative, and desensitisation to trauma triggers in a safe therapeutic environment (Yule, Smith, & Perrin, 2005). In particular, sand-play therapy has been found clinically useful for children as a medium for processing abuse and violence trauma in children (Grubbs, 1994; Parson, 1997). In this way, the therapist helps the child to learn that they can approach their fears and not experience the feared consequences, leading to an overall reduction in anxiety and trauma symptoms in everyday life.

As highlighted, a contextual framework is particularly important when working with children, adolescents, and families, particularly where abuse has occurred. The CBT model considers a child's thoughts, feelings and behaviours not only in relation to specific situations but also in relation to his or her broader context (Graham, 2005; Reinecke, Dattilio, & Freeman, 2003). That is, the TF-CBT program is designed to assist the therapist to take into account the finer nuances of antecedents and consequences of the child's presenting problems, as well as the interactive effects of his or her history and development, family relationships and attachments, social and cultural context, symptom expression and response to therapy. The approach assumes that a comprehensive assessment has been carried out prior to treatment beginning, and involves an ever-evolving formulation and ongoing consultation with parents, caregivers and other professionals.

In developing treatments, outcome testing is essential. Thus, before dissemination can occur the program must be evaluated. Being a development project, the current research first involved a single-case design, as recommended (Kazdin, 2000; King & Ollendick, 2006) and used by clinical researchers for the initial testing of a manualised treatment protocol (e.g., Girling-Butcher & Ronan, 2009; Kane & Kendall, 1989; Lees & Ronan, 2008). Single-case designs have also been recommended as particularly useful for the conduct of research in applied clinical settings in a managed care environment (Morgan & Morgan, 2003), and are consistent with a scientist–practitioner approach (Feather, 2004).

Thus, a single-case multiple-baseline design across participants involving both intra- and inter-participant replications (Hersen & Barlow, 1976) was used to demonstrate the effects of the TF-CBT treatment on PTSD symptoms and child coping. Child comorbid symptoms and parent/caregiver and teacher views on child functioning were also assessed at selected intervals. The research program to date has been conducted over four studies, designed to build on each other (Feather, 2007). Each study has involved four multiply abused children aged 9–15 years who met diagnostic criteria for PTSD, and their parent/caregivers. The first phase of the research involved the development and piloting of the TF-CBT program. Reported here, the second phase of the research was designed to evaluate the fully developed manualised TF-CBT protocol. These evaluations were carried out in a community setting with agency clinicians and typically referred children.

To briefly summarise the first phase of the research, Feather and Ronan (2006) piloted the protocol with four Pakeha/New Zealand European children, the most commonly referred population group to the agency. The findings showed that participants reported an overall reduction in PTSD symptoms and an increase in coping behaviours related to specific abuse and trauma-related concerns, compared to pre-treatment levels. Generally, these gains were maintained over follow-up intervals (3 months, 6 months and 12 months). An additional study by Feather, Ronan, Murupaenga, Berking, and Crellin (2009) piloted the protocol across cultures, with two Maori and two Samoan children, representing the minority cultural groups most often referred to this New Zealand agency. The main purpose of that study was to ascertain whether the program could be adapted to be culturally appropriate and meet the needs of these children and families, and result in effective outcomes. Participants also showed an overall reduction in PTSD symptoms and an increase in coping that was maintained over follow-up intervals (Feather et al., 2009). During this initial development phase, the senior author delivered the therapy.

The purpose of the current phase of the research program was to evaluate the fully developed TF-CBT program across different therapists, all of whom were seeing typically referred patients in a specialist clinic of New Zealand's statutory child protection agency. Study 1 involved evaluating use of the manualised protocol, with the senior author as therapist. Study 2 involved a replication study, evaluating use of the manualised protocol by two other agency therapists. In both studies the children were typically referred with multiple abuse histories and current PTSD diagnoses, representing a diverse range of cultures. It was expected that participants would show a reduction in PTSD and related symptoms and an increase in coping that would be maintained over follow-up intervals.

Method

Participants

Study 1 participants were two boys aged 13 years (P1.2, P1.3), and two girls aged 11 (P1.1) and 12 years (P1.4). Study 2 participants were two boys aged 11 (P2.3) and 12 years (P2.2), and two girls aged 9 (P2.4) and 12 years (P2.1). All children met *Diagnostic and statistical manual of mental disorders* (4th ed., DSM-IV) (American Psychiatric Association, 1994) diagnostic criteria for PTSD as the primary diagnosis. All presented with multiple abuse histories, including physical abuse, sexual abuse, neglect, emotional abuse, and witnessing domestic violence. Across both studies, the children and families involved reflected the diverse range of cultural backgrounds of children typically referred to the agency, including Pakeha/New Zealand European, Maori, Samoan, Eastern European, North African and South American.

There were no exclusion criteria specified explicitly for the research because all referrals were subject to the agency's criteria. All children referred to the agency were "open cases" with New Zealand's statutory child protection service. Children typically presented with a mixed picture of abuse, trauma, child protection and child behaviour concerns. Children who may not have been accepted included those who presented primarily with child mental health concerns, who were more appropriately referred to a child mental health service; or children with primarily a sexual abuse history who may, according to New Zealand statute, be referred for counselling funded by the Accident Compensation Corporation. The majority of children and parents/caregivers invited to participate in the research agreed. Over all four studies, of the 24 approached, 22 agreed to take part and 16 finished the program (Feather, 2007). Details on those in Studies 1 and 2 who did not finish are now provided. A 10-year-old Pakeha girl (Study 1) was withdrawn from the sample early in the treatment phase because clinical judgement indicated that she required another treatment approach, due to aspects of developmental delay. A 10-year-old Maori girl (Study 2) did not complete the program due to lack of caregiver support for therapy. In each study the next child referred to the agency who met research criteria was included.

Measures

A multi-method, multi-informant battery of instruments was used, as recommended by the child maltreatment and child trauma literature (American Academy of Child and Adolescent Psychiatry, 1998; Myers et al., 2002). A full battery of self-report measures, and parent/caregiver and teacher measures was designed to assess the impact and symptomatology of child abuse trauma before, after and at 3-month and 6-month follow-up. All the measures selected have known reliability and validity and have been shown to be sensitive to the effects of treatment.

The current research was primarily focused on answering questions related to selected instruments from this battery that tracked ongoing progress. In particular, in keeping with a single-case design and scientist–practitioner model of service delivery (Feather, 2004; Kazdin & Nock, 2003), the continuous self-report measures completed by the children that tracked changes in PTSD symptoms (Children's Post-Traumatic Stress Reaction Index [CPTS-RI]) and self-identified coping factors (Coping Questionnaire [CQ]) were of particular interest.

Anxiety Disorders Interview Schedule for Children. The Anxiety Disorders Interview Schedule for Children (ADIS) (Silverman & Albano, 1996) is a structured clinical interview administered to children, with a parallel version for parents. Revisions have ensured that it is compatible with the DSM-IV and can be used for diagnostic purposes (Silverman, 1994). The ADIS has been used extensively by childhood anxiety researchers due to its high reliability and user-friendly structure, which enhances rapport and enables in-depth gathering of information on the child's historical and current functioning (Ronan, 1996).

Children's Post-traumatic Stress Reaction Index. The CPTS-RI (Frederick, Pynoos, & Nader, 1992) is a 20-item measure rated on a 5-point Likert scale (scored 0–4) that assesses the features and symptoms of PTSD in children. Widely used in research, it can be administered in a self-report or interview format. Scores can range from 0 to 60, with a cut-off of 12+ indicating a mild PTSD reaction; 25+ a moderate PTSD reaction; and 40+ a severe PTSD reaction.

State Trait Anxiety Inventory for Children. The State Trait Anxiety Inventory for Children (STAIC) (Spielberger, 1973) measures both acute (state) and chronic (trait) anxiety in children aged 9–12 years. Each scale has 20 items rated on a 3-point Likert scale (scored 1, 2, or 3). The Trait scale is treatment sensitive and can be used to measure the effectiveness of clinical procedures designed to reduce anxiety in children. Scores can range from 20 to 60, with higher scores indicating greater anxiety.

Children's Depression Inventory. The Children's Depression Inventory (CDI) (Kovacs, 1981) assesses affective, behavioural and cognitive signs of depression. It has 27 items. Each item has three choices (scored 0, 1, or 2), each of which characterises the child over the previous 2 weeks. It is designed to measure the severity of depressive symptoms in children and adolescents aged 8–17 years. Scores can range from 0 to 54, with higher scores indicating greater severity.

Coping Questionnaire. The CQ-C (Kendall et al., 1992) measures the self-perceived coping ability of a child in specific anxiety-provoking situations. The three situations most distressing for the child are identified during the assessment interviews and are listed on the CQ-C and rated each week by the child on a 7-point Likert scale ranging from *not at all able to help myself* (1) to *completely able to help myself feel comfortable* (7).

Child Behaviour Checklist/4-18–Parent Form. The Child Behaviour Checklist/4-18–Parent Form (CBCL) (Achenbach, 1991a) measures parent/caregiver reports of child competencies, emotional functioning and behaviour problems in a standardised format. Two sections measure (a) social competencies, assessing the child's participation in areas such as sports, hobbies, and social interactions, and (b) problematic behaviours, consisting of 118 problem items rated on a 3-point scale (scored 0, 1, or 2). The problem item raw scores are recorded on a scoring profile, which provides percentile and *T* scores based on norms for each sex in each age range, presented in a series of problem scales.

Child Behaviour Checklist–Teacher Report Form. The Child Behaviour Checklist–Teacher Report Form (TRF) (Achenbach, 1991b) is used to assess child problems and competencies in the realm of social/emotional functioning at school. It mirrors the parent version of the CBCL and provides teacher ratings of the child's academic performance, adaptive functioning, and behavioural/emotional problems.

Research design

A single-case multiple-baseline across participants involving intra- and inter-participant replications (Hersen & Barlow, 1976) was used to demonstrate the controlling effects of treatment on PTSD symptoms and child coping. Repeated measures were used to assess child comorbid symptoms and parent/caregiver and teacher views on child functioning. In each study, the four participants were randomly assigned to a baseline of 3, 5, 7, or 9 weeks at intake. Continuous (weekly) measures were completed throughout the baseline and treatment phases. As in the first two studies (Feather & Ronan, 2006; Feather et al., 2009), a non-concurrent design element was used (Hayes, 1981; Watson & Workman, 1981). This adjusts and relaxes the need for participants to begin assessment concurrently. It allows for using data across several participants seen at different times. Consistent with a multiple-baseline design, it is intended to exploit baseline intervals of variable duration. Multiple-baseline designs themselves are intended to provide experimental evidence for intervention effects and to reduce the possibility that change occurs as a function of other processes, including placebo effects linked to both assessment and intervention (Blampied, 1999; Kazdin, 2003). The criterion of course is successive replications across the variable of interest (i.e., in these studies, across participants). To be more specific, one quite specific aim of the current study was to assess for change in PTSD symptoms and coping ability as a function of the introduction of treatment compared to variable baseline conditions. In addition to the evaluation of the impact of phased treatment elements, this design, through its use of ongoing assessment, can also account for any documented external events (Kazdin & Nock, 2003).

Procedure

Following referral, an assessment interview with a trained independent assessor was scheduled with the child and parent/caregiver. If the child met diagnostic criteria for PTSD, informed consent from the child and legally responsible adult(s) was sought. The full battery of assessment measures was completed with the child, parent/caregiver and teacher. The child was randomly assigned to a baseline duration of 3, 5, 7 or 9 weeks. The child completed the weekly measures of the short assessment battery (CPTS-RI and CQ) during the baseline period and during treatment. The full battery was administered again after treatment and at 3-month and 6-month follow-up points. The ADIS was not re-administered after treatment or at follow-up, due to the length of time it took to administer and the extent of assessment demands that had already been made on the participants. As per the informed consent agreement, if the child did not wish to complete measures at any time this was respected. If there had been concerns that the research constraints would impact on safety, clinical concerns would have been prioritised.

For Study 1, therapy was provided by the senior author (a developer of the program), then a doctoral candidate, and an experienced child and family clinician. For Study 2, therapy was provided by two clinic staff who were trained in using the manualised TF-CBT program by the senior author. The assessments were administered by a trained and independent assessor, a postgraduate clinical psychology student, except when she was not available.

Treatment manual

The TF-CBT treatment program is set out in a 16-session format, with regular scheduled parent/caregiver sessions (Feather & Ronan, 2004). For a fuller description, see also the pilot study report (Feather & Ronan, 2006). The manual is designed to be flexible to cater for the range of issues that may present, and to be adapted to different cultural contexts and developmental stages. To summarise the approach here, the treatment is phased, with each phase emphasising important elements identified in the literature, complemented with aspects from local clinical practice. Phase 1 focuses on relationship and contextual factors, in particular, psychosocial strengthening and social support. Phase 2 is derived from the CBT program for anxiety disordered children developed by Kendall et al. (1990), and helps the child develop coping skills based on a four-step coping template, "The STAR Plan". Phase 3 is based on a gradual exposure procedure for processing and resolving trauma, derived from behavioural, cognitive and expressive therapy models adapted for children (Macdonald, Lambie, & Simmonds, 1995; Yule, Smith, & Perrin, 2005). Phase 4 provides a transition to life beyond therapy; any issues that are so far unresolved are addressed and relapse prevention is covered. Booster sessions are offered if required.

Results

The results of Study 1 and Study 2, comprising the second phase of the research program are presented.

Study 1 results

Treatment fidelity. A senior clinical psychologist (and experienced cognitive behavioural therapist) independent from the current research reviewed 13% of randomly selected audiotapes of treatment sessions with the two children who consented to this procedure, in accordance with recommendations that an acceptable practice is to use three sessions per patient (Waltz, Addis, Koerner, & Jacobson, 1993). Adherence to treatment content and session goals was rated 100% for delivery, and no alternate treatment strategies were identified.

Variations from research design. The assessments for Study 1 participants were conducted by a trained evaluator not involved in the treatment. Owing to the assessor leaving the agency, however, the researcher (and senior author) carried out the follow-up assessments.

It was planned that the baseline data would be collected weekly for 3, 5, 7, and 9 weeks. As found in the previous two studies, participants experienced unforeseen delays between the initial assessment and planned treatment commencement for a range of reasons, including placement changes, caregiver unavailability, and transport difficulties (total weeks on baseline: P1.1, 7; P1.2, 9; P1.3, 14; P1.4, 17). In addition, three children failed to return some baseline measures (P1.2, 2; P1.3, 2; P1.4, 4). The baseline data reported here are those of the measures returned over the prescribed baseline period (in order of the dates recorded on the coversheets, and not necessarily representing consecutive weeks).

Length of treatment, booster sessions, and follow-up data. The 16-session TF-CBT program was delivered flexibly, as described in the manual. Number of treatment sessions varied from 11 to 17, and length of treatment ranged from 15 to 33 weeks (P1.1, 17 sessions, 33 weeks; P1.2, 11 sessions, 15 weeks; P1.3, 15 sessions, 25 weeks; P1.4, 16 sessions, 18 weeks). The differential treatment requirements of participants are explained in following sections. Duration of treatment was primarily affected by placement changes and caregiver availability.

Booster sessions were requested for one child by her social worker periodically over the 9 months following the completion of the 16-session program (P1.1; 16 additional sessions, 33 in total). Coping skills were reviewed and reinforced to address issues related to changes in placement and contact with family.

Follow-up data were available for two participants at 3 and 6 months (P1.1, P1.3), and one participant at 3 months only (P1.4). Following discharge from child protection involvement, P1.4 was not able to be located at 6 months, and P1.2 did not consent to complete follow-up measures.

Caregiver involvement. Caregiver involvement varied depending on individual circumstances. Due to repeated changes of caregiver, continued caregiver involvement for one child (P1.1) was not possible. In the case of the two adolescent boys (P1.2, P1.4), their respective non-offending parents had issues of their own related to the abuse. This was resolved by referring each parent to an agency therapist to enable them to address their own issues concurrently. The parent/caregiver elements of the TF-CBT program were covered flexibly by the child therapist and the parent therapist, as appropriate. P1.3's caregiver brought her to each session, and participated fully in the program as prescribed.

Child reports: Continuous measures. The overall mean results of the child self-report measures that tracked changes in PTSD symptoms (CPTS-RI) and self-identified coping factors (CQ) of particular interest to this research are presented in Figures 1, 2.

Figure 1 shows that the level of PTSD symptoms of Study 1 participants decreased with treatment, and for three participants with data available, decreased further over a 6-month follow-up period. The overall average self-reported CPTS-RI score for the four children was in the upper mild range during baseline ($M = 22.6$, $SD = 12.0$), just below clinical levels over the treatment phase ($M = 9.4$, $SD = 9.5$), and well below clinical levels at follow-up ($M = 3.5$, $SD = 3.3$). Figure 2 shows that Study 1 participants' overall level of coping increased with treatment and increased further over the following 6 months for three participants. The children's mean coping scores (averaged across the four participants' three self-identified target concerns; CQ1, CQ2, CQ3) increased from 4.5 ($SD = 1.3$) during the baseline phase, to 5.7 ($SD = 0.8$) over treatment; and 6.3 ($SD = 0.8$) over follow-up for three participants.

Figure 3 presents the weekly level of PTSD symptoms on the CPTS-RI for each of the four participants across the baseline, treatment and follow-up phases. Study 1 participants generally showed a downward trend in PTSD symptoms over the three phases, with gains maintained at follow-up. The results of each phase are analysed more specifically using visual inspection, quantitative,

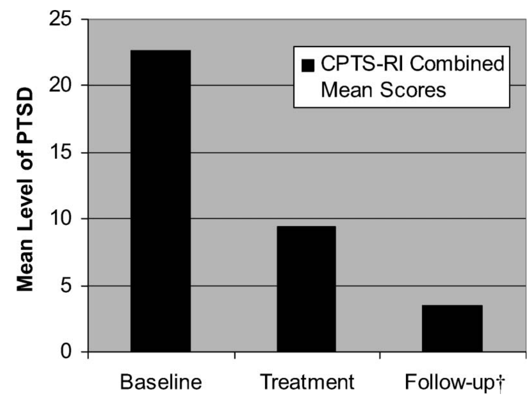

Figure 1. Study 1: changes in mean level of post-traumatic stress disorder (PTSD) symptoms (average of Children's Post-Traumatic Stress Reaction Index [CPTS-RI] scores for all four participants). †Three children, five data points.

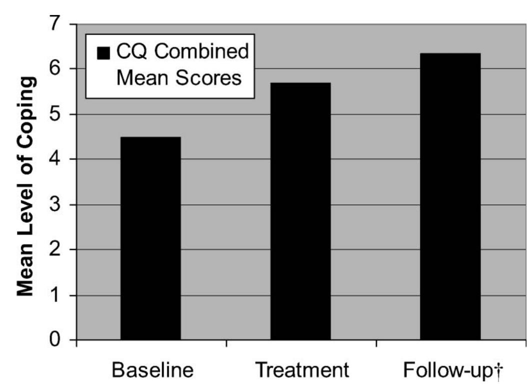

Figure 2. Study 2: changes in mean level of child self-reported coping (average of Coping Questionnaire [CQ] scores for all four participants). †Three children, five data points.

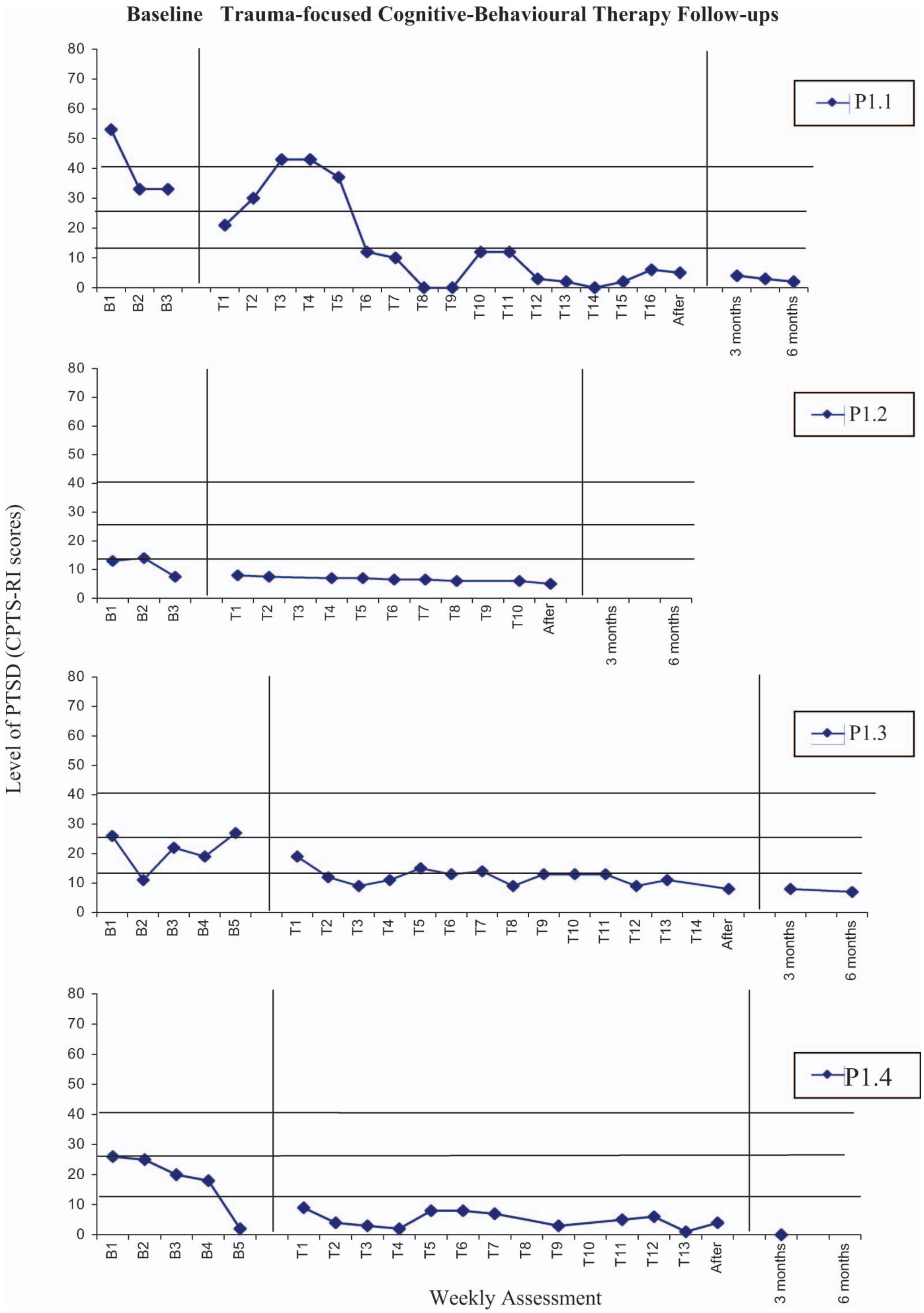

Figure 3. Study 1: changes in post-traumatic stress disorder (PTSD) symptoms (Children's Post-Traumatic Stress Reaction Index [CPTS-RI] scores).

and qualitative data, as recommended in the single-case literature (Kazdin, 1982).

A stable baseline demonstrates little variability and an absence of slope. For the CPTS-RI, the individual baseline ranges and variability were 33–53 (25%), 8–13 (6.3%), 11–27 (20%), and 2–26 (30%), for P1.1, P1.2, P1.3, and P1.4 respectively. Although no participant exceeded the 50% level recommended for single-case methodology in applied clinical research (Kazdin, 2003), three participants (P1.1, P1.2, P1.4) showed a downward trend in CPTS-RI scores following the initial assessment. While P1.1 recorded a reduction in PTSD symptoms following the initial assessment, her baseline scores stabilised at moderate levels, warranting TF-CBT intervention. Conversely, the baseline data for P1.2 and P1.4 dropped below clinical levels, and called into question the need for trauma-focused treatment. These results are discussed with reference to clinical aspects.

The baseline results for P1.2 show CPTS-RI scores in the mild range, although re-experiencing, avoidance, and hyper-arousal symptoms at initial assessment were reported, and collateral data confirmed sufficient symptoms at levels warranting a PTSD diagnosis. Notably, as part of the forensic interviewing process, P1.2 had already given an account of the serious domestic violence he had witnessed. This process appears to have been therapeutic, functioning as exposure. In addition, P1.2 underwent preparation for court during the baseline phase. Due to his involvement in these activities, P1.2 expressed an initial reluctance to attend therapy, possibly contributing to underresponding. His mother, however, continued to be concerned about his wellbeing and behaviour. Given these clinical concerns, a decision was made to commence treatment with Phase 1 and 2 of the TF-CBT program, and reassess the need for the exposure phase for treating PTSD.

P1.4 recorded a moderate CPTS-RI score at initial assessment. During the ADIS-C interview he made significant new disclosures to the assessor about past sexual abuse. A number of factors are likely to have contributed to a subsequent drop in PTSD symptoms during the baseline phase. As noted, the process of disclosure can be therapeutic. The disclosures necessitated the development of a relationship with P1.4 by the assessor to ensure his safety, according to agency protocol. Notably, P1.4 was Samoan. The salience of the therapeutic relationship for Samoan children and ensuing symptom reduction was noted in the cultural pilot study (Feather et al., 2009). Consultation with caregiving adults and overall clinical concerns led to a decision to commence treatment with Phase 1 and 2 of the TF-CBT program, and reassess the need for the exposure phase for treating PTSD.

Visual inspection of the CPTS-RI scores over the treatment phase indicates a similar pattern of a gradual reduction and maintenance of symptoms to below clinical levels for all but one participant, who showed an initial rise in symptoms, followed by a reduction (P1.1).

Clinical observations provide a fuller picture of the data as follows. P1.1 experienced a traumatic placement breakdown in the early weeks of therapy. She was subsequently placed in a safe, supportive caregiving environment.

During treatment, P1.2's CPTS-RI scores remained below clinical levels throughout the psychosocial strengthening and coping skills phases of the program. Given that he was no longer reporting PTSD symptoms, and collateral reports indicated that the other clinical concerns had abated, a decision was made to exclude the exposure phase and complete treatment after Phase 2 with relapse prevention.

Similar to P1.2, P1.4's CPTS-RI scores remained below clinical levels throughout the treatment phase. It was agreed that P1.4 would work through salient incidents of maltreatment he had experienced using the trauma processing modalities available, even though his PTSD symptoms had abated. Follow-up CPTS-RI results show that PTSD symptoms remained below clinical levels for the participants for whom data was available at 3-month (P1.1, P1.3, P1.4) and 6-month (P1.1, P1.3) assessment points.

Figure 4 presents each child's self-reported coping over the baseline, treatment and follow-up phases, as measured on the CQ. The baseline data for the CQ scores is relatively stable in terms of variability and trend. Overall, child coping showed a gradual upward trend over the treatment phase and was maintained at overall higher levels at 3- and 6-month follow-up for the participants for whom data were available.

Individual baseline ranges and variability on the CQ were 4.0–4.0 (0%), 3.8–5.3 (21%), 3.0–4.3 (19%), and 5.7–7.0 (19%) for P1.1, P1.2, P1.3, P1.4, respectively, all within acceptable limits for single-case research. Two participants had a stable or slightly downward trend (P1.1, P1.3), while P1.2 had an initial downward trend, spiking upwards prior to treatment commencing, and P1.4 had a gradual upward trend over the baseline phase, indicating an increase in coping. These results correspond to those presented for the CPTS-RI. That is, those participants who recorded a decrease in PTSD symptoms over baseline, showed a related increase in coping.

The graphed results during the therapy phase reflect individual patterns of response to treatment. The individual results for P1.1 show an increase in coping linked with the beginning therapy and a slight increase over the psychosocial phase of treatment. She recorded a marked drop just prior to the coping skills phase, related to an incident of stealing. Her coping resumed throughout the coping skills training, and increased markedly during the gradual exposure phase, when she worked through early traumatic experiences. P1.1's coping was maintained at high levels, with a slight drop associated with ending therapy, as can sometimes happen for children as they anticipate coping without the support of therapy.

For P1.2, the beginning of treatment coincided with being called to court as a witness. This was, of course, the focus of therapy at the outset, flexibly

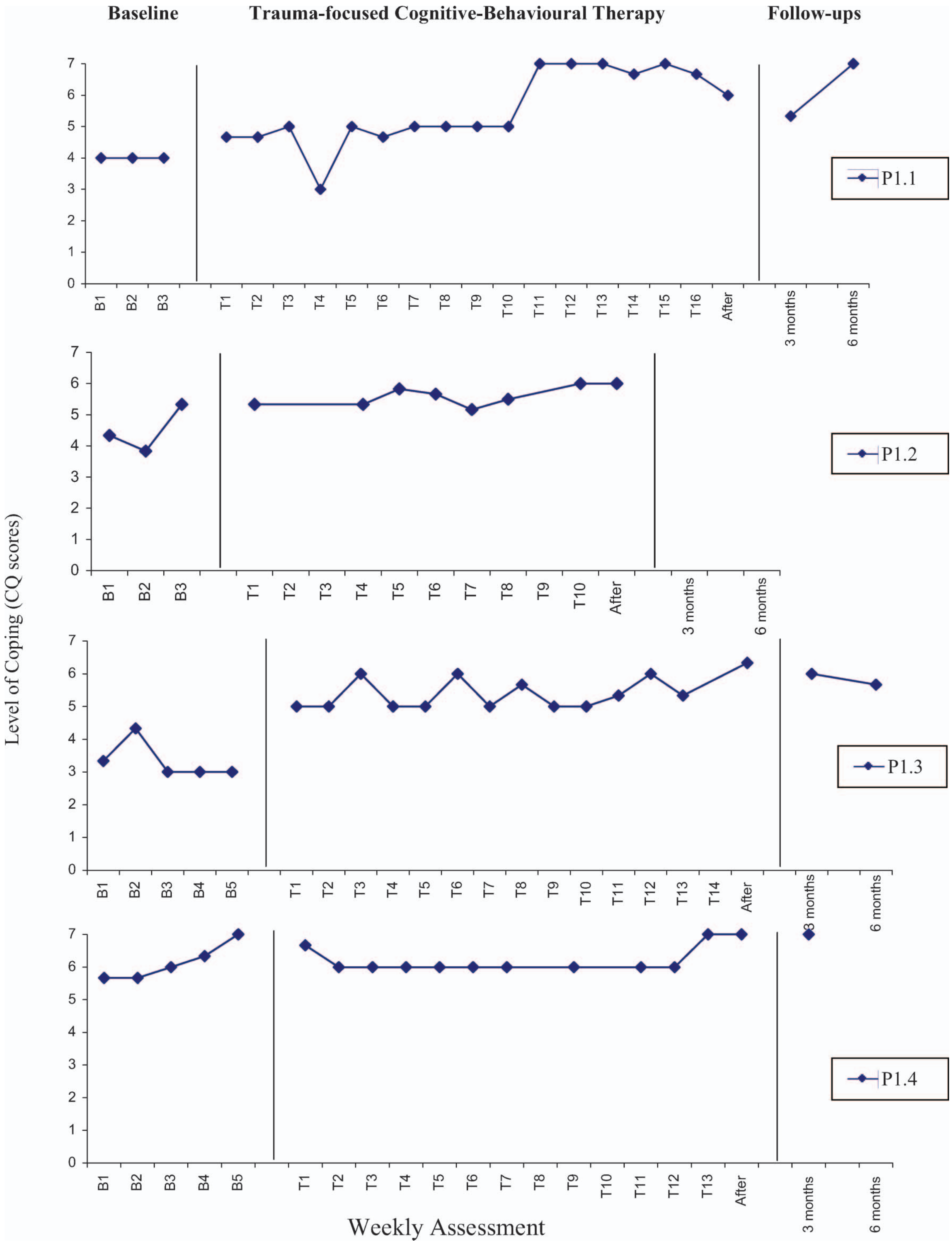

Figure 4. Study 1: Changes in child reported coping skills (average of scores for three target concerns). CQ = Coping Questionnaire.

utilising TF-CBT program elements as required to assist the young person to cope with this process. The results show that P1.2's coping gradually increased and was maintained at high levels by the end of therapy. The exposure component was not necessary, and treatment was concluded with relapse prevention.

P1.3's coping increased at the outset of treatment compared with baseline levels. Her coping was maintained at high levels with minor fluctuations over the first two phases of treatment, and continued to gradually increase over the exposure phase.

P1.4's coping showed a slight drop at the outset of treatment, but maintained stability at a high level throughout Phases 1–3, with an increase in coping to maximum levels by the end of treatment.

The follow-up CQ results for three participants indicate that self-reported coping was maintained (P1.4), or dropped slightly (P1.1, P1.3), at 3 months; and for two participants, either continued to increase (P1.1) or maintained at a similar level (P1.3).

Child reports: Repeated measures. Child self-report scores related to target concerns and comorbid problems are presented in Table 1. These data fill out the symptomatology picture for the participants at pre- and post-treatment and follow-up assessment points.

At pre-treatment assessment the mean CPTS-RI score was in the moderate range ($M = 29.5$). One participant recorded a score above clinical cut-off for severe PTSD (P1.1), two for moderate PTSD (P1.3, P1.4), and one for mild PTSD (P1.2). At post-treatment assessment all participants scored below clinical cut-off (range = 4–8, $M = 5.5$), with treatment gains maintained at 3-month follow-up for three participants (P1.1, P1.3, P1.4; range = 0–8, $M = 6.0$), and 6 months for two participants (P1.1, P1.3; range = 2–7, $M = 4.5$).

The mean CQ score for all four participants increased from before treatment ($M = 4.3$, on a scale of 1–7) to after treatment ($M = 6.3$). For participants for whom data were available, mean CQ was maintained at 3-month ($M = 6.1$), and 6-month ($M = 6.2$) follow-up assessment points.

Scores on the STAIC-T show that trait anxiety was within the normal range for all four participants at initial assessment, reduced further at post-treatment assessment, and was maintained at similar levels at follow-up for each of the three participants for whom data were available.

CDI scores were below clinical cut-offs for depression for all participants at all assessment points. Three participants had a reduction in scores at post-treatment assessment compared to pre-treatment assessment (P1.1, P1.2, P1.4), and one participant maintained scores at a similar low level (P1.3). Available follow-up data indicated that CDI scores were maintained at similar levels to post-treatment assessment at 3 months (P1.1, P1.3. P1.4) and 6 months (P1.1, P1.3).

The opportunity for children to provide subjective feedback on their experience of participation in the program was provided throughout the study. The feedback provided by P1.1 is reported here, as an example. At the outset of treatment, P1.1 noted that she would like help with "Being less upset, coping with school, coping at home". At 3-month follow-up, P1.1 reported, "I'm more calm, less fidgety. I'm more easier to talk with. I lie and steal less. I am able to stop myself from lying and stealing and I think of the consequences".

Table 1. Study 1: Child report scores on repeated measures of target and comorbid symptoms

		Assessment points			
Measure	Participant	Before treatment	After treatment	3-month follow-up	6-month follow-up
CQ (Mean)	P1.1	4.0	6.0	5.3	7.0
	P1.2	4.3	6.0	–	–
	P1.3	3.3	6.3	6.0	5.3
	P1.4	5.7	7.0	7.0	–
	Mean	**4.3**	**6.3**	**6.1**	**6.2**
CPTS-RI (Total)	P1.1	53‡	5	4	2
	P1.2	13	5	–	–
	P1.3	26†	8	8	7
	P1.4	26†	4	0	–
	Mean	**29.5**	**5.5**	**6.0**	**4.5**
STAIC-T (*T* score)	P1.1	58	48	46	41
	P1.2	35	21	–	–
	P1.3	38	25	25	27
	P1.4	44	24	27	–
	Mean	**43.8**	**29.8**	**32.7**	**34.0**
CDI (*T* score)	P1.1	57	44	47	44
	P1.2	44	37	–	–
	P1.3	36	38	36	36
	P1.4	48	44	46	–
	Mean	**46.3**	**40.1**	**43.0**	**40.0**

Notes. CDI = Child Depression Inventory; CPTS-RI = Child Post-traumatic Stress Reaction Index; CQ = Coping Questionnaire; PTSD, post-traumatic stress disorder; STAIC-T = State Trait Anxiety Inventory for Children–Trait.
†Clinical cut-off for moderate PTSD, CPTS-RI ≥25; ‡Severe PTSD, CPTS-RI ≥40. – = data not returned or incomplete.
Bold = mean scores.

With regard to program elements, P1.1 reported that the following had been particularly helpful:

> *The relaxation thingies – the breathing. I use it whenever I'm angry. The fact that there's help. Sand trays – being able to express my feelings so I don't have to keep it in me any more, so someone else knows. The STAR Plan – helped when it was needed – would probably still use it if I really needed to.*

The only aspect P1.1 identified that had not been helpful was the timing of therapy, which meant she had missed out on school. When asked what she thought had helped the most, P1.1 stated, "All of it. My attitude has changed – I'm more calm."

Parent, caregiver and teacher reports. For Study 1, the parent/caregiver CBCL and teacher CBCL-TRF data were considered too inconsistent to provide meaningful results. Factors that contributed to inconsistency included changes in caregivers (P1.1), changes in teachers (P1.1, P1.3, P1.4), parent(s) with cultural world view different from test developers, and English as a second language (P1.2, P1.4). Linked to these issues, some measures were not completed, despite every endeavour made (CBCL, P1.2, P1.4; TRF, P1.3). Due to these difficulties a decision was made not to pursue follow-up parent/caregiver and teacher quantitative measures for Study 1, but to gather qualitative data when possible. These data reflect the individual responses of children to the treatment program, and views of the respective adults involved in their lives, and are available from the authors on request.

Study 2 results

Treatment fidelity. Treatment fidelity was conducted on 30% of the audiotapes of treatment sessions with the one child in Study 2 who consented to this procedure. Adherence to treatment content and session goals was rated 100% for delivery, and no alternate treatment strategies were identified.

Variations to research design. The participants were randomly assigned to weekly baseline lengths of 3, 5, 7, and 9 weeks. Three participants experienced delays between the initial assessment and planned treatment commencement (total weeks on baseline: P2.1, 13; P2.2, 15; P2.3, 23; P2.4, 9). The reasons here were similar to those of participants in previous studies. In addition, two children failed to return two sets of baseline measures each (P2.2, P2.3). The baseline data on the graphs represent the measures collected over the requisite weeks of the baseline phase for each participant (data points on Figures 5,6 are in order of dates on the coversheets, and do not necessarily represent consecutive weeks).

Length of treatment and assessment data. As with the previous studies, the 16-session TF-CBT program was delivered flexibly, as described in the manual. Number of treatment sessions varied from 9 to 25, delivered over time periods ranging from 18 to 68 weeks (P2.1, 17 weeks, 41 weeks; P2.2, 17 sessions, 34 weeks; P2.3, 9 sessions, 18 weeks; P2.4, 25 sessions, 68 weeks). Extended treatment phases were related to a range of reasons.

In terms of assessment data, one participant requested not to complete measures at six assessment points during treatment (P2.4: T13, T14, T17, T18, T19, T20), and at 3-month follow-up. This was of course respected. Follow-up data were available for one participant at 3 months (P2.3) and two participants at 6 months (P2.2, P2.4). Although every endeavour was made, follow-up measures were not returned by three participants who had moved out of the area (P2.1, P2.2) or had been discharged from child protection involvement (P2.3).

Caregiver involvement. The parent/caregivers of all participants were involved in the assessment and treatment. There were variations and limitations, however, to involvement due to a range of factors, such as caregiver availability and placement changes.

Cultural factors. It is of note that all children and their parent/caregivers involved in Study 2 were of cultural minorities (including Maori, Samoan and a recent migrant family). All strongly identified with their culture and, with the exception of the Maori caregivers, English was their second language. These cultural factors were addressed in a number of ways. The two Samoan children and their families were allocated to the Samoan therapist. The therapist of the other two families consulted liberally and as required with his Maori supervisor with regard to P2.2, and with a colleague of relevant cultural heritage with regard to P2.3.

On the basis of the Feather et al. (2009) study the researcher guided the therapists to prioritise the world view of their clients, and to use the TF-CBT as a tool-kit flexibly, as intended, taking care to include all treatment elements. With regard to the use of Western-based measures, given that the children were acculturated into the New Zealand Western-based education system and English was their first language, it was considered appropriate to use the child measures. The parent/caregivers, however, were less acculturated, and English was generally not their first language. While endeavours were made to administer the parent/caregiver measures at the outset, if the preference was to give verbal feedback, this was respected.

Child reports: Continuous measures. Figure 5 shows that the level of PTSD symptoms of Study 2 participants, as measured on the CPTS-RI, decreased with treatment, and for three participants with data available, decreased further over a 6-month follow-up period. The overall average self-reported CPTS-RI score for the four children was in the upper mild range during baseline ($M = 23.2$, $SD = 6.8$), in the mid-mild range over the treatment phase ($M = 19.7$, $SD = 8.8$), and below clinical levels for the participants for whom data were available at follow-up ($M = 9$, $SD = 4.6$).

Figure 6 shows that the overall level of Study 2 participants' self-reported coping increased with treatment, as measured on the CQ. Coping increased further over the following 6 months for three participants. The children's mean CQ scores increased from 4.8 ($SD = 1.3$) during the baseline phase, to 5.8 ($SD = 1.2$) over treatment; and 7.0 ($SD = 0.0$) over follow-up for three participants.

Figure 7 presents the graphed weekly level of PTSD symptoms for each of the four participants across the baseline, treatment and follow-up phases.

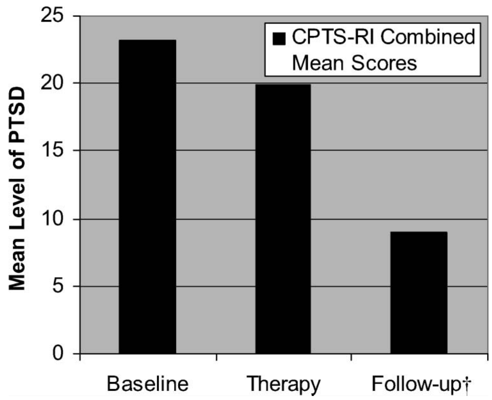

Figure 5. Study 2: changes in mean level of post-traumatic stress disorder (PTSD) symptoms (average of Children's Post-Traumatic Stress Reaction Index [CPTS-RI] scores for all four participants). †Three children, three data points.

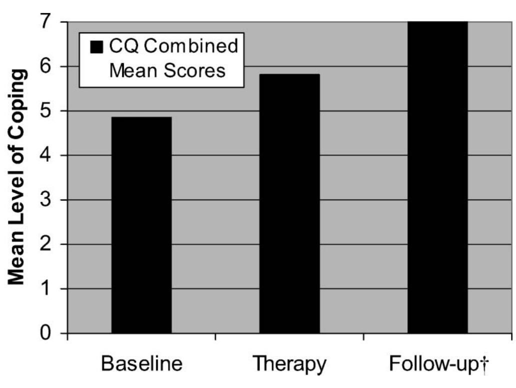

Figure 6. Study 2: changes in mean level of child self-reported coping (average of Coping Questionnaire [CQ] scores for all four participants). †Three children, three data points.

Study 2 participants had relatively little individual variability in baseline CPTS-RI scores, but some evidence of downward slope following initial assessment. Available follow-up data indicated gains maintained below pre-treatment levels. The results are discussed using visual inspection and relevant qualitative data, as recommended in the single-case literature (Kazdin, 1982).

The CPTS-RI baselines of all four participants demonstrated variability well within the acceptable limits for single-case methodology in applied clinical research (Kazdin, 2003). The individual baseline ranges and variability were 24–33 (11%), 23–25 (3%), 6–21 (19%), and 23–35 (15%), for P2.1, P2.2, P2.3, and P2.4, respectively. Three participants showed a downward trend immediately following initial assessment (P2.1, P2.3, P2.4), although only one had scores that continued to trend downwards (P2.3). Visual inspection of the CPTS-RI scores over the treatment phase indicates idiosyncratic patterns of response for Study 2 participants. These are reported with reference to qualitative data.

P2.1 showed a drop in PTSD symptoms associated with beginning therapy, apparently linked with developing a therapeutic relationship with her Samoan therapist. Her symptoms, however, subsequently increased, linked with ongoing family conflict and neglect issues. P2.1 was ultimately placed in out-of-home care, and began to make treatment gains. By post-treatment assessment, P2.1's CPTS-RI score had reduced to below clinical levels.

P2.2's CPTS-RI scores were in the moderate range during the early phases of treatment, with the exception of a reported reduction in PTSD symptoms to below clinical levels, coinciding with the end of the coping skills phase. His scores, however, subsequently rose and remained elevated in the moderate range throughout the remainder of treatment. P2.2 later disclosed that he had been getting physical abused in his placement. This is likely to explain the rise and ongoing elevation of his PTSD symptoms.

At the beginning of treatment, P2.3's reported CPTS-RI scores in the mild range, reducing to below clinical levels by the end of the coping skills phase. At this point clinical concerns had abated and it was decided to discontinue treatment. A slight spike in PTSD symptoms, back into the mild range, was noted at post-treatment assessment, as can occur for children as they contemplate coping on their own without the support of therapy.

P2.4's CPTS-RI scores showed an initial drop, coinciding with engagement with her Samoan therapist, demonstrating a similar pattern to P2.1. Her scores fluctuated in the mild range over the initial phases of treatment, during which time she was living with extended family. An increase in

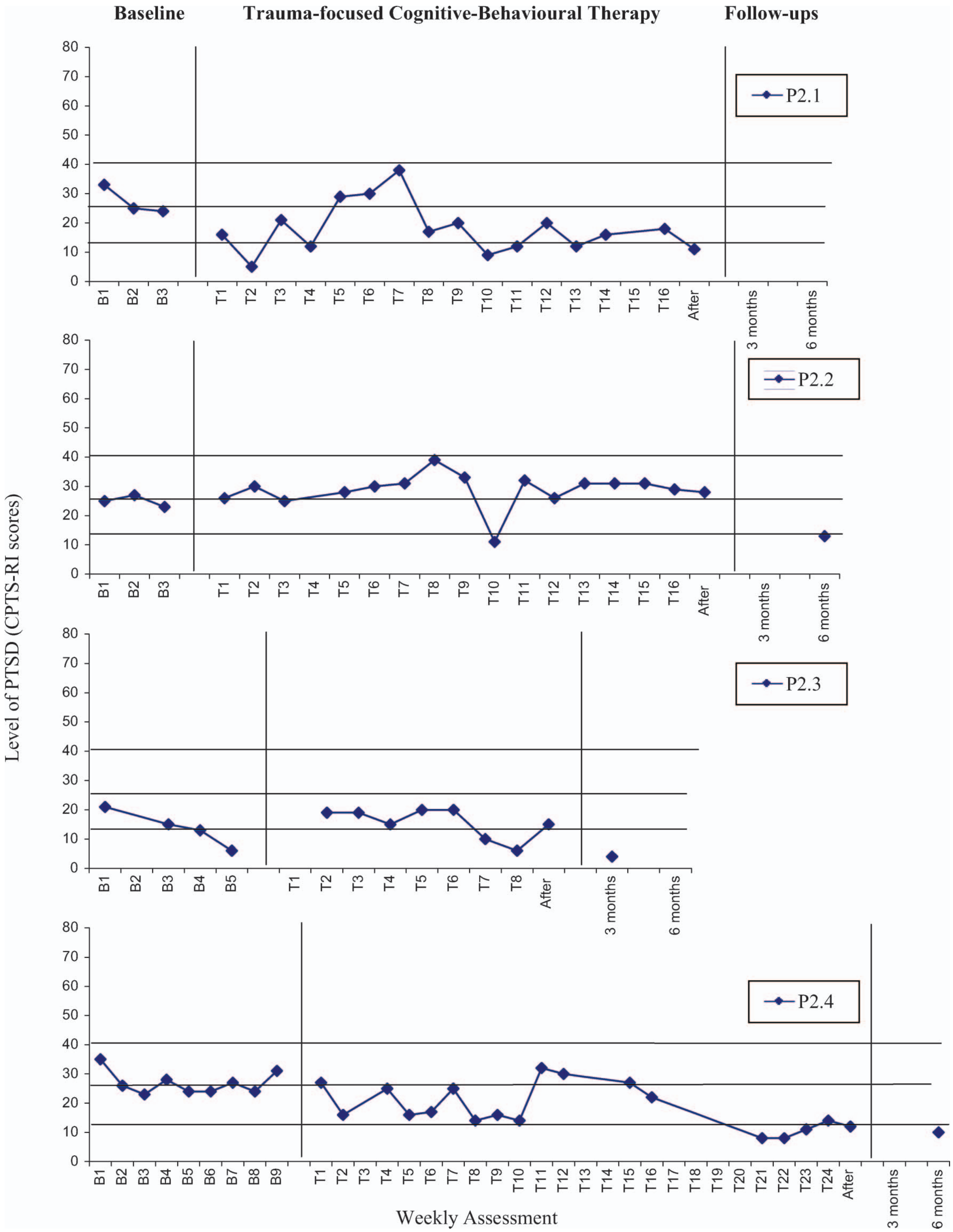

Figure 7. Study 2: changes in post-traumatic stress disorder (PTSD) symptoms (Children's Post-Traumatic Stress Reaction Index [CPTS-RI] scores).

scores to the moderate range during the exposure phase (T11) was associated with the outcome of a family meeting. A clinical decision was made to suspend the planned exposure sessions and address boundary and safety issues with individual parent and family sessions. Over this period, P2.4's scores remained in the moderate range. Once the family sessions were completed, it was agreed she would resume the TF-CBT program (T15). She was now able to process her past abuse trauma via a series of sand trays, and her CPTS-RI scores dropped to below clinical/borderline mild.

Available follow-up data show that for three participants, PTSD symptoms decreased further, to below clinical levels (P2.3, 3-month follow-up; P2.4, 6-month follow-up) or borderline mild (P2.2, 6-month follow-up).

Figure 8 presents each child's self-reported coping over the baseline, treatment, and follow-up phases. For three participants, the baseline CQ data were stable in terms of variability and trend. One participant (P2.4) reported an increase in coping after the initial assessment, and subsequent stability at a high level. Over the treatment phase, child coping showed an upward trend, or stability at a high level. Available follow-up data indicated maintenance of treatment gains at high levels of coping.

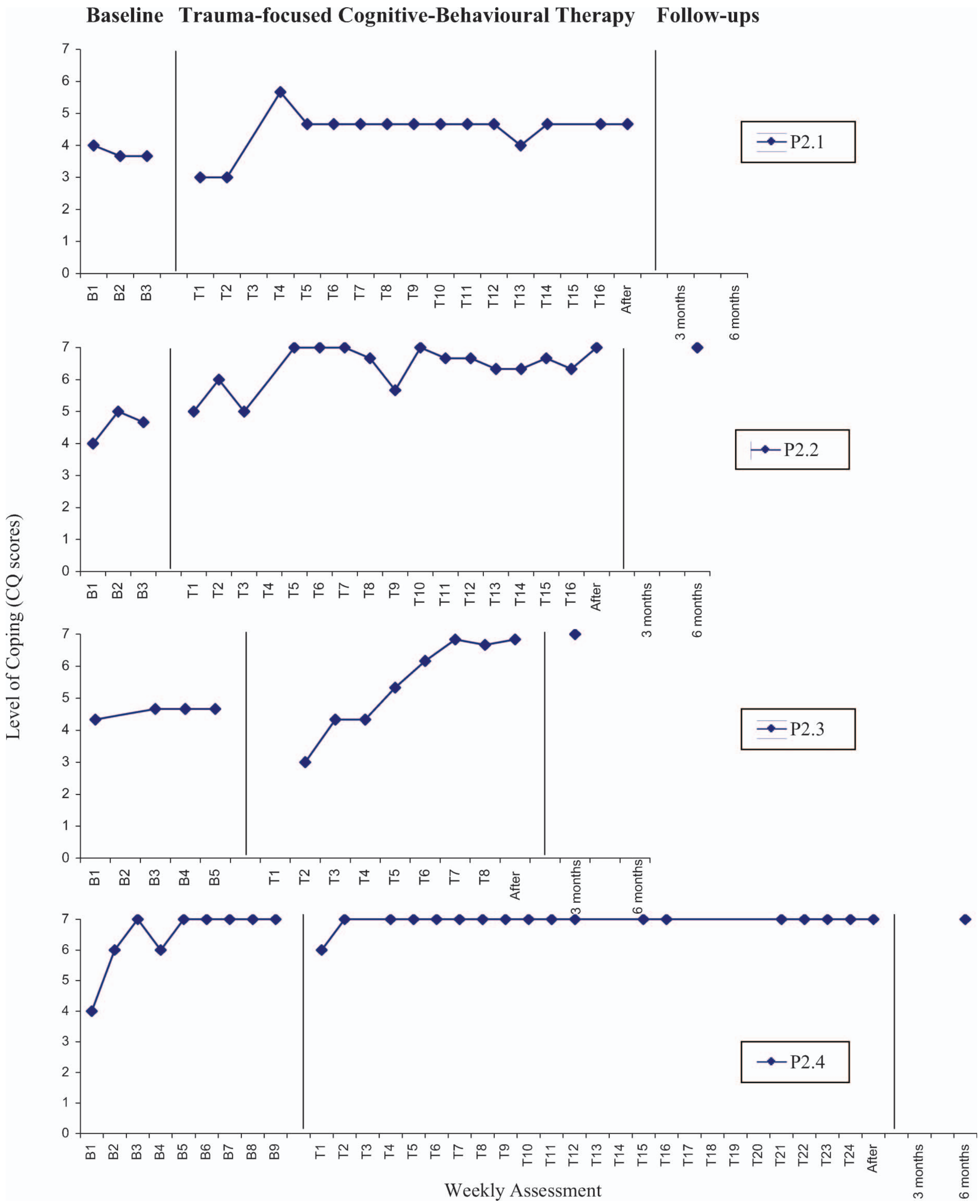

Figure 8. Study 2: changes in child reported coping skills (average of scores for three target concerns). CQ = Coping Questionnaire.

For the CQ, the individual baseline ranges and variability were within acceptable limits: 3.7–4.0 (4%), 4.6–5.0 (6%), 4.3–4.7 (1%), and 4.0–7.0 (43%), for P2.1, P2.2, P2.3, P2.4, respectively. Three participants demonstrated a stable baseline with regards to trend, while one participant demonstrated an increase in coping following the initial assessment (P2.4). This was the youngest Samoan child and, as discussed, this may reflect the salience of engagement and instilled hope on her self-perceived ability to cope.

Notably, all four participants in Study 2 continued to demonstrate an increase in coping with treatment, with the exception of P2.2, who achieved and maintained maximum coping scores during baseline that were maintained across therapy and at 6-month follow-up (Figure 8). For the three participants for whom follow-up CQ data were available, self-reported coping was maintained at highest levels at 3 months (P2.3), and 6 months (P2.2, P2.4).

Child reports: Repeated measures. Child self-report scores related to target concerns and comorbid problems are presented in Table 2. These data fill out the symptomatology picture for participants at pre- and post-treatment, and follow-up assessment points.

At pre-treatment assessment the mean CPTS-RI score for Study 2 participants was in the moderate range ($M = 28.5$). Three participants reported scores above clinical cut-off for moderate PTSD (P2.1, P2.2, P2.4), and one for mild PTSD (P2.3). At post-treatment assessment P2.2 reported PTSD in the low moderate range and the other three participants reported scores in the low mild range (P2.3, P2.4), or below clinical cut-off (P2.1). At follow-up the three participants for whom data were available reported a further reduction in PTSD symptoms to below clinical cut-off (P2.3, 3-month follow-up; P2.4, 6-month follow-up) or low mild (P2.2, 6-month follow-up).

The mean CQ score for all participants increased from before treatment ($M = 4.1$) to after treatment ($M = 6.4$), and for participants for whom data were available, increased further at follow-up assessment points ($M = 7.0$) on a scale of 1–7.

Scores on the STAIC-T show that trait anxiety was in the normal range for all participants at pre- and post-treatment assessment, and was maintained at similar levels at follow-up for each of the three participants for whom data were available.

CDI scores were below clinical cut-offs compared to age and gender norms for all participants at all assessment points. A reduction of scores from pre-treatment levels is noted.

Table 2. Study 2: Child report scores on repeated measures of target and comorbid symptoms

		Assessment points			
Measure	Participant	Before treatment	After treatment	3-month follow-up	6-month follow-up
CQ (Mean)	P2.1	4.0	4.7	–	–
	P2.2	4.0	7.0	–	7.0
	P2.3	4.3	6.8	7.0	–
	P2.4	4.0	7.0	–	7.0
	Mean	**4.1**	**6.4**	**7.0**	**7.0**
CPTS-RI (Total)	P2.1	33†	11	–	–
	P2.2	25†	28†	–	13
	P2.3	21	15	4	–
	P2.4	35†	12	–	10
	Mean	**28.5**	**16.5**	**4.0**	**11.5**
STAIC-T (*T* score)	P2.1	53	21	–	–
	P2.2	48	49	–	48
	P2.3	40	24	21	–
	P2.4	45	21	–	25
	Mean	**46.5**	**28.7**	**21**	**36.5**
CDI (*T* score)	P2.1	49	36	–	–
	P2.2	60	38	–	46
	P2.3	42	–	37	–
	P2.4	40	37	–	39
	Mean	**47.7**	**37.0**	**37.0**	**42.5**

Notes. CDI = Child Depression Inventory; CPTS-RI = Child Post-traumatic Stress Reaction Index; CQ = Coping Questionnaire; PTSD, post-traumatic stress disorder; STAIC-T = State Trait Anxiety Inventory for Children–Trait.
†Clinical cut-off for moderate PTSD, CPTS-RI ≥ 25. – = data not returned or incomplete.
Bold = mean scores.

Child subjective data were collected, and are available from the authors, if required.

Parent, caregiver and teacher reports. For Study 2, due to incomplete or unreturned measures from parents and caregivers, partly due to changes in caregiving arrangements (P2.1, P2.2, P2.4), or lack of consent (P2.3), CBCL reports were considered too inconsistent to be valid. These data and subjective parent and caregiver reports are also available from the authors.

Discussion

A manualised TF-CBT program was developed based on relevant theory, empirical research, local practice and contextual factors (Feather & Ronan, 2004). To date, it has been evaluated over two separate pilot studies with maltreated children diagnosed with PTSD at referral to a specialist clinic of the statutory child protection agency in New Zealand. The current research was intended to evaluate the capacity of the final version of the treatment protocol delivered by (a) a developer of the program and (b) other therapists not involved with development.

The findings support the effectiveness of the TF-CBT program delivered by both the developer and by different therapists. Given the single-case research design used in this research, the design itself relies upon both intra- and inter-participant replications (Morgan & Morgan, 2003). Inter-participant replications showed that levels of PTSD symptoms reduced and child coping increased across baseline and treatment. Intra-participant findings were also supportive and improvements on most indicators were noted across treatment. That is, ongoing assessment across baseline and treatment demonstrated a similar trend of individual reduction in symptoms and increases in coping as a function of intervention. Over follow-up, where data were available, this trend continued: change was generally either maintained or continued to improve.

The fact that the majority of these children continued to demonstrate relief from symptoms and an increase in coping as a result of the treatment elements is consistent with previous treatment outcome research, which has shown the long-term effectiveness of CBT for anxiety-related problems in children (Kendall & Southam-Gerow, 1996), and the usefulness of systematic desensitisation and exposure in resolving trauma-related symptomatology in children (Farrell, Haines, & Davies, 1998; Saigh, 1987) and adults (Foa & Rothbaum, 1998; Foa, Rothbaum, Riggs, & Murdock, 1991). These current findings support the effectiveness of a TF-CBT for maltreated traumatised children and report outcomes similar to those reported for comparable intervention models developed in the United States (e.g., Cohen, Mannarino, & Deblinger, 2006) and consistent with a body of research emerging internationally showing that TF-CBT is superior to other treatments in improving PTSD and related symptoms in multiply traumatised school-age children (Cohen, Mannarino, Murray, & Igelman, 2006; Wethington et al., 2008).

Positive treatment outcomes of the current studies demonstrate not only the effectiveness of the manualised TF-CBT approach with therapists newly trained in the approach, but also that treatment gains can be clinically significant for children and their families representing a range of multiple-abuse histories, cultural backgrounds and different caregiving arrangements. Cultural issues were carefully addressed, drawing on the learning from our cultural pilot study (Feather et al., 2009). In addition, treatment integrity was established. Importantly, children made and sustained gains based on an intervention carried out in a clinic setting within an agency specialising in working with serious child maltreatment cases. In contrast, there were a number of limitations, with some of these limitations related to the difficulties of this work in real-life settings.

Limitations

Limitations of the current research include some variations to the initial design that included unforeseen changes to baseline duration, and missing baseline and follow-up data. Additionally, as with one of the pilot studies (Feather et al., 2009), treatment length was extended for some participants, which increases the likelihood of the effect of maturation, history, multiple treatment interference or other extraneous variables on the outcome results (Kazdin, 2003). In addition, the original plan to collect 12-month follow-up data had to be abandoned due to restrictions on the data collection period. For 3- and 6-month follow-up, data were not able to be collected, for a variety of reasons documented and linked to real-life issues impinging on a systematic protocol. A notable limitation of the findings was that, due to agency constraints, the therapists on occasion administered and collected the assessment measures. This may have influenced results in a direction more favourable to the research hypotheses, due to demand characteristics and reactivity of experimental arrangements (Kazdin). In addition, in one way, it was a limitation of Study 1 that these patients were seen by the developer, a therapist with intimate knowledge and high allegiance to the protocol. It was for this reason that use of the manual by other therapists was carried out in Study 2. When generalising from the results, however, it needs to be kept in mind that this sample was small, with only two therapists having delivered the program to two children each.

Additional limitations to the validity of the results from the quantitative measures were identified. In particular, participants of cultural backgrounds and world views different to that of the test developers expressed difficulties in understanding questions on the measures or reluctance to complete measures, with unknown effects on the results (Andary, Stolk, & Klimidid, 2003). Furthermore, there were considerable missing data on adult measures for a range of reasons, including frequent changes in caregiving arrangements and schooling. Where data were available, it seemed to vary depending on how well the adult knew the child, whether they relied on what the child told them, or even their own agenda, which at times seemed to lead to underreporting or overemphasising child symptoms. These problems with the use of adult measures in child outcome studies are similar to those noted by previous researchers in this area (Davis & Siegel, 2000; Kendall-Tackett, Williams, & Finkelhor, 1993).

As with the pilot studies, the current studies are limited by methodological problems related to carrying out research in a day-to-day clinic setting within a child protection statutory agency. These include resource capacity, and, importantly, the need to prioritise care and protection concerns and patient best interests over the research protocol. In addition, psychological treatment at the clinic is concurrent with the social work intervention, with unknown multiple-treatment interference.

Finally, the fact that findings for this protocol are based overall on a series of single-case evaluations (see also Feather & Ronan, 2006; Feather et al., 2009), means that more research is necessary with larger groups to confirm the potential of this intervention. Having said that, across four different evaluations, including the two studies reported here, and despite limitations, preliminary support is provided, including findings linked to clinical significance, that do justify continuing development of this approach.

Implications

Taken together, the results of all four studies to date show that children varied in the number of treatment sessions required to complete the program (9–28 individual child sessions) and, relatedly, to demonstrate treatment responses (up to 45 face-to-face contacts, including child and parent/caregiver booster sessions and case worker support related to implementation of the TF-CBT program). One quarter of the participants required booster sessions. Broadly, this finding is convergent with that of other researchers in the area of child abuse trauma, who have similarly found that some children require as many as 40 or more sessions depending on the needs of the child and the complexity of the case (Deblinger & Heflin, 1996). Consistent with other studies on CBT for anxiety and abuse and trauma-related problems, treatment response appeared to vary with developmental/age level (Cohen, Berliner, & March, 2000; Kane & Kendall, 1989). In the case of the current research, the results across the four studies to date suggest that younger children (e.g., P2.4, age 9) may take longer to integrate the coping skills into their everyday lives than adolescents.

Longer length of treatment also appeared to relate to contextual factors, particularly safety. Not surprisingly, when children were exposed to situations in which their safety was at risk, whether physical or emotional, trauma symptoms were frequently re-triggered or remained elevated, and treatment gains stalled or were compromised. Notably, once these children were placed in a safe environment and/or safety concerns were resolved, progress in therapy then appeared to resume.

With regard to parent/caregiver involvement, the children who had continuity of support throughout the assessment and treatment demonstrated a response to treatment that was more rapid and lasting. This is consistent with previous research in this area, in which parent support has been found to predict treatment outcome for children with PTSD as a result of abuse (Cohen & Mannarino, 2000). Additionally, in the area of conduct disorder in youth, recent New Zealand research has highlighted the central role of family in facilitating and supporting positive treatment outcomes in youth (Curtis, Ronan, Heiblum & Crellin, 2009; Ronan & Curtis, 2008; Russell & Ronan, 2009). In this research, parent/caregiver involvement varied considerably, as would be expected in a care and protection clinic setting. Every endeavour was made to involve non-offending parents and/or caregivers in the treatment. This was achieved by administering this aspect of the intervention flexibly when necessary, for example, before and after child sessions and by telephone. Overall, it appeared that the salient factor was a supportive caregiver who was willing to be involved in the treatment, whether or not a parent. Again, however, this more anecdotal finding requires more systematic data support.

Treatment response patterns are of interest to researchers and clinicians alike. Although the overall results for all four studies showed similar overall patterns, the single-case design enables the idiosyncratic responses to treatment phases by individual participants to be discerned. Participants demonstrated a range of responses to the psychosocial strengthening phase. The coping skills phase was often associated with an improvement in PTSD symptoms and an increase in coping. For many children, the exposure phase appeared to bring about a temporary increase in symptoms while past trauma was processed (e.g., P1.3, P2.1, P2.2, P2.4). Ultimately, treatment gains were generally maintained or continued to improve (e.g., P1.1, P1.3, P1.4, P2.2, P2.4).

The treatment response of a number of participants suggested that, regardless of the complexity and severity of the abuse history, if the child was able to process and resolve unhelpful abuse-related attributions and perceptions (e.g., transferring blame to someone other than the perpetrator; P2.4) as well as engage in exposure-based processing, treatment gains were made. Once again, this appeared to be age-related, because clinical observations suggested that adolescents seemed to be able to process unhelpful attributions more readily than younger children. These findings suggest that age, current cognitions and related affect are particularly salient, and support those of other maltreatment researchers, who have found that abuse-related attributions may predict treatment response more strongly than abuse-related factors such as identity of the perpetrator, abuse type, or number of abusive episodes (Cohen & Mannarino, 2000). Mindful of the single-case design used, more research would be worthwhile to produce a more differentiated picture and to assess whether the other factors suggested here do in fact reliably predict treatment response in this population.

With regard to cultural minorities, previous research has found that culture appears to be less significant in predicting response to treatment than other factors, such as those described above (Cohen, Deblinger, Mannarino, & Arellano, 2001). The current research bears this out. Preliminary findings indicated that the TF-CBT program was helpful for children and families of minority indigenous and migrant cultures in New Zealand, provided that it was delivered in the context of a culturally sensitive collaborative therapeutic relationship and flexibly adapted to the clients' world view. This is consistent with current understanding that the important factors in engaging cross-culturally are flexibility of approach and respect of cultural perspectives (Banks et al., 2006).

With regards to the assessment measures used in the research, although multimodal multi-source evaluations are recommended in the child abuse and child trauma literature in order to obtain corroborating information, response rate and consistency were problematic in this research due to the nature of the clinical population. For example, more than half of the children experienced at least one change of caregivers and three quarters had at least one change of school during the research period. For two fifths of parent/caregivers, English was their second language. The current research highlights the need for researchers and clinicians to be aware of multiple factors, including potential biases, in pursuing multimodal, multi-source assessment when evaluating a young person for abuse and trauma and measuring treatment progress and outcome. More recently developed assessment measures, such as the new Achenbach System of Empirically Based Assessment (ASEBA) (Achenbach & Rescorla, 2007), which includes cross-informant comparisons and multicultural norms, may address some of these problems. This system is currently being normed internationally. The development of local measures, however, may ultimately be equally beneficial.

Importantly, the TF-CBT treatment package was generally perceived as acceptable and helpful by the children and families involved, as indicated by anecdotal reports of children and parents/caregivers. Additional quantitative child reports showed that the majority of children found that the TF-CBT approach helped them to feel less upset while they were engaged in treatment, and these gains were typically maintained at follow-up for the children for whom data were available (Feather, 2007). Clearly, acceptability and perceived helpfulness are linked with effectiveness in terms of treatment adherence and cooperation, as well as being a fundamental obligation for clinicians and treatment providers (King & Ollendick, 2006).

Finally, it could be argued that this project was perhaps too ambitious given the population and clinical setting. The pattern of results, however, across all four studies suggests that useful research outcomes can be obtained by local clinicians treating complex cases in real-life settings using single-case research designs.

Future directions

Given the supportive initial findings for this TF-CBT program, a next step is to do evaluation using controlled and randomised designs (efficacy studies) as well as further studies that mirror conditions seen in usual practice (effectiveness studies). Suggestions for future research that arise from the current research include (a) development and evaluation of the TF-CBT protocol for treating related populations of children who may present with trauma primarily from sources other than abuse, but where abuse may also have occurred (e.g., refugee children and families); (b) further development and evaluation of TF-CBT treatment elements targeting particular developmental/age groups, including children younger than in the current study; (c) further development on the interweaving of cultural models and TF-CBT, and evaluation with cultural minorities (see also Feather et al., 2009); (d) further research on development and evaluation of culturally sensitive outcome measures for this population, and evaluation of the new measurement system with cross-informant comparison and multicultural norms, and development of local norms (i.e., ASEBA; Achenbach & Rescorla, 2007); and (e) randomised controlled trials in at least two settings, which would be required in order to determine whether the TF-CBT protocol is "efficacious" (additionally, effectiveness studies using group comparison designs are similarly necessary). Of course, these future directions rest on current and previous research (Feather & Ronan, 2006; Feather et al., 2009) that itself rests on the philosophy and values of a "local scientist" tradition. Thus, replication of the TF-CBT program using various design strategies, including pragmatic single-case designs, in practice settings in Australasia where maltreated traumatised children and families are referred is warranted. This would include child mental health services, non-governmental agencies and private clinics, particularly those that receive referrals from child protection services.

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